



Project Report -2

Wazuh SIEM File
monitoring and log
analysis



Prepared By:
ABDULLAH-AL-KAYES

Objective : Monitoring Files and Directory activity
using FIM module.

Date : 05/10/2026

01 Environment

- Host : VMware Workstation
- OS : Ubuntu 24.04 LTS
- Wazuh-Version: 4.14 All-in-One



02. STEP-BY-STEP DEPLOYMENT

1. Enabled File Integrity Monitoring (FIM) :

- Edited wazuh agent configuration:

```
sudo bash  
sudo nano /var/ossec/etc/ossec.conf
```

- Added directories to monitor under the <syscheck> section:

```
<directories
```

```
check_all="yes"realtime="yes">/home/hunter/documents</directories>
```

- Restarted the Agent for the config to take effect :
`sudo systemctl restart wazuh-agent`

2. Generated activity for testing :

- Created, modified and deleted files in monitored directories.
- Installed software packages using “sudo apt” to trigger file changes alerts.
- performed various file operations to simulate real-world activity.

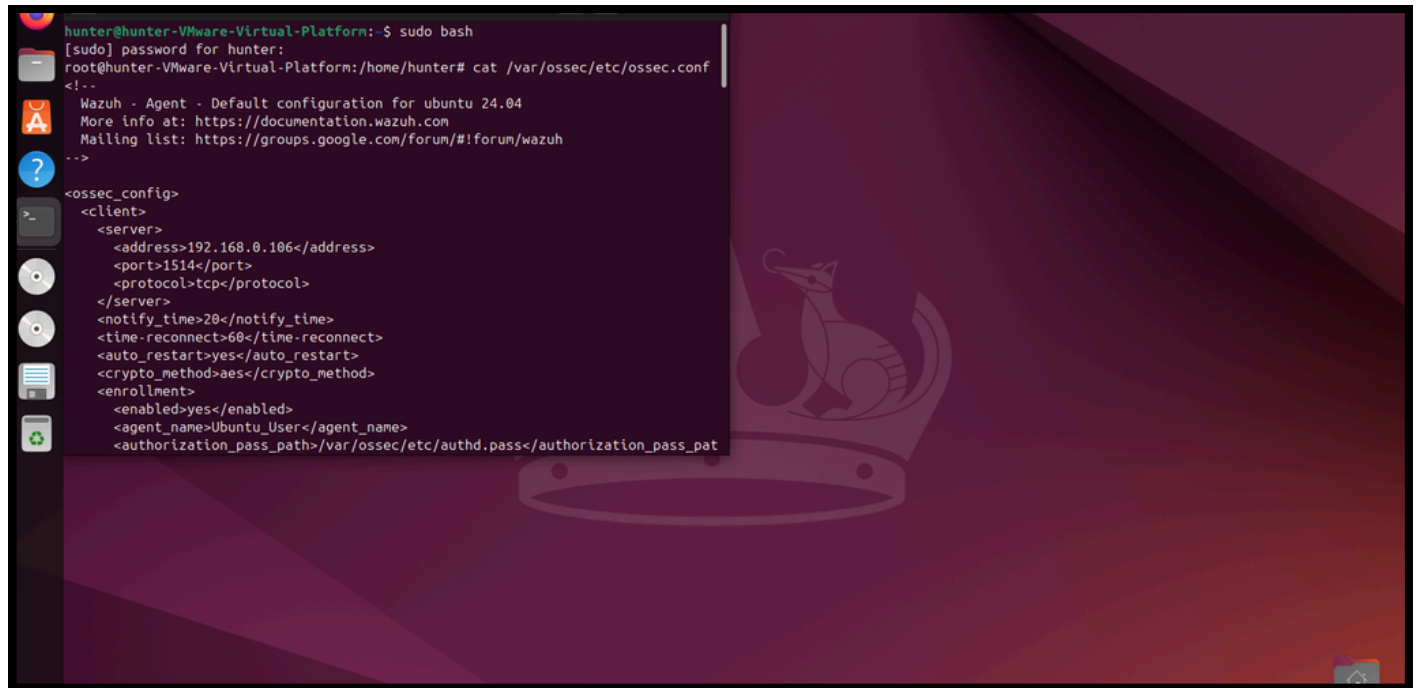
3. Dashboard Analysis :

- Explored Security Events. Agents and Vulnerabilities tabs in Wazuh agent dashboard.
- Filtered alerts by rule ID, agent name, and FIM data source.
- Investigated 6+ alerts in detail (file added, modified, and deleted events).



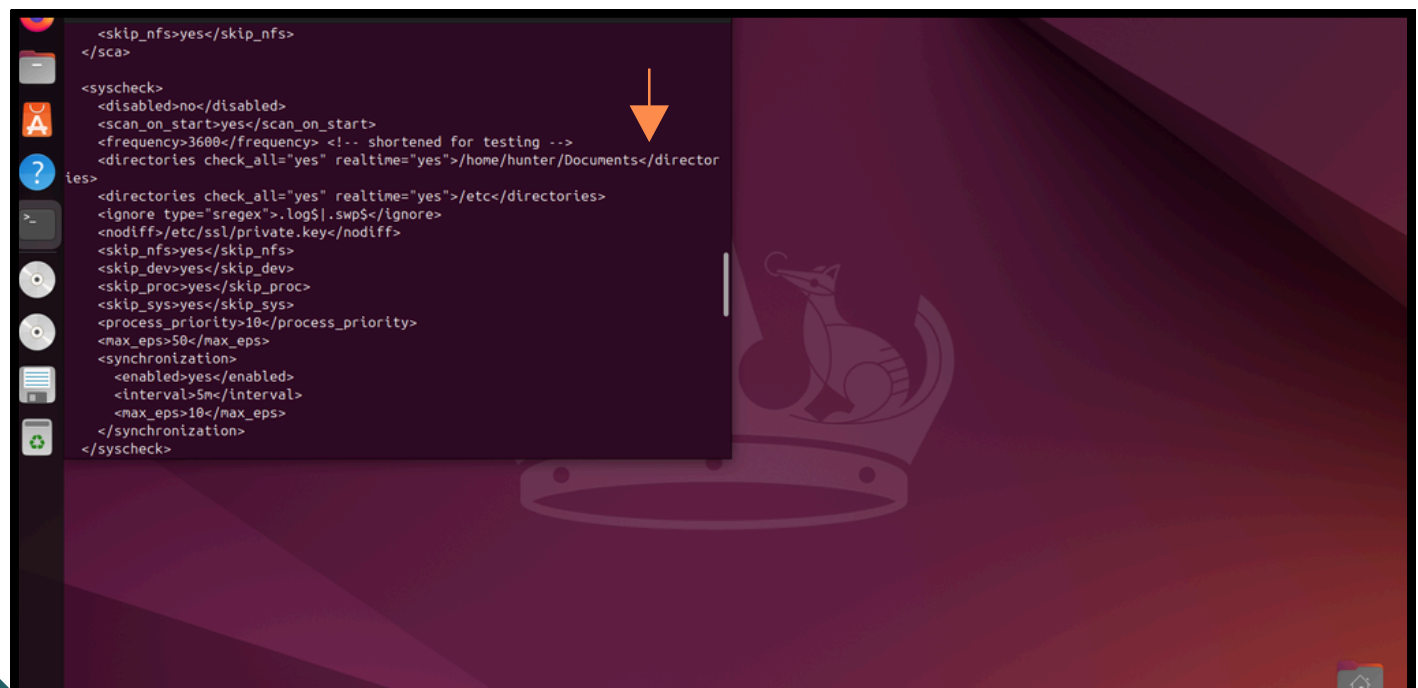
02 Screenshots

Screenshot 01 : FIM configuration on ossec.conf



```
hunter@hunter-VMware-Virtual-Platform:~$ sudo bash
[sudo] password for hunter:
root@hunter-VMware-Virtual-Platform:/home/hunter# cat /var/ossec/etc/ossec.conf
<!--
Wazuh - Agent - Default configuration for ubuntu 24.04
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>
  <client>
    <server>
      <address>192.168.0.106</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <notify_time>20</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
    <enrollment>
      <enabled>yes</enabled>
      <agent_name>Ubuntu_User</agent_name>
      <authorization_pass_path>/var/ossec/etc/authd.pass</authorization_pass_pat
```



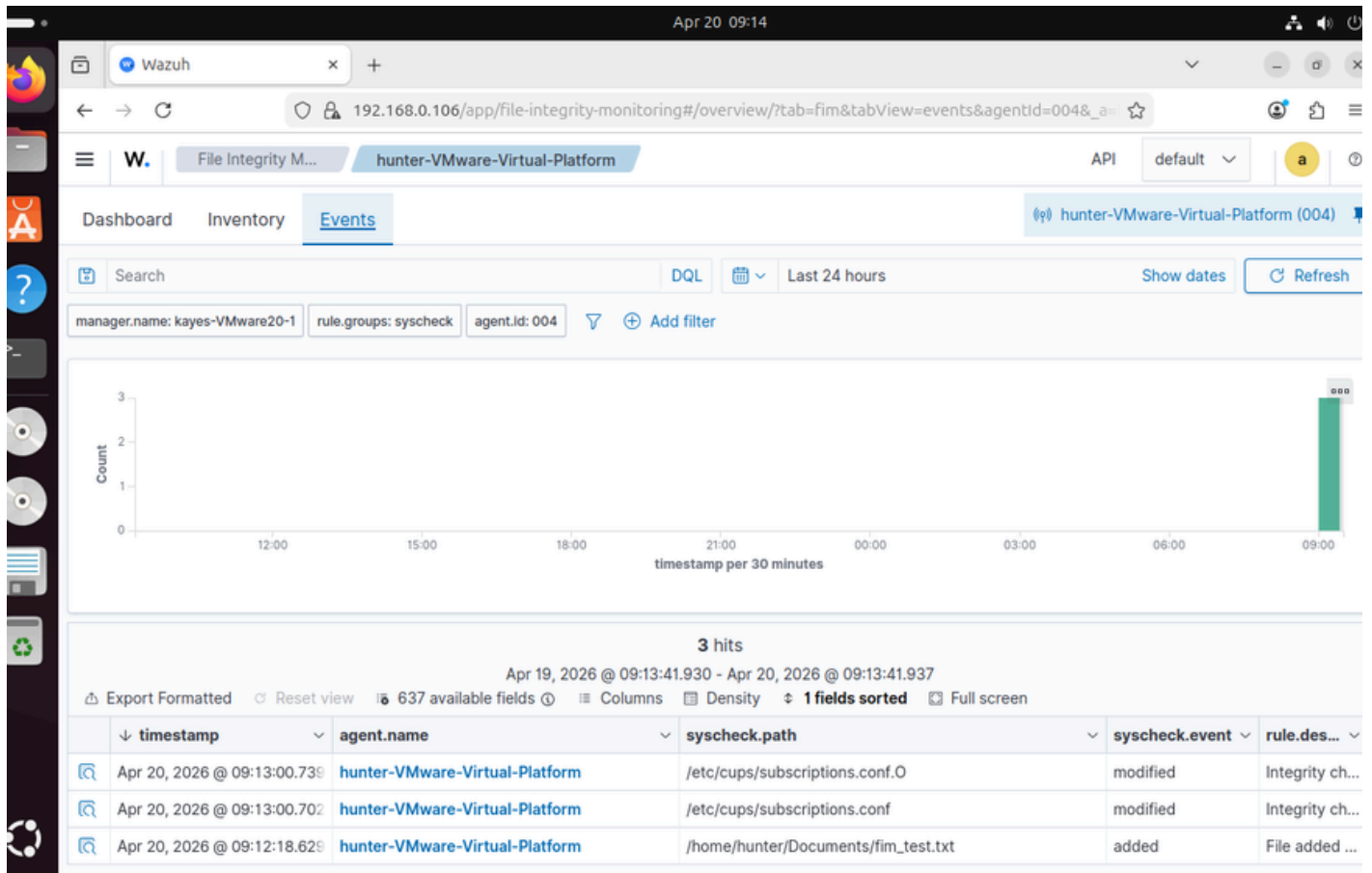
```

    <skip_nfs>yes</skip_nfs>
  </sca>

  <syscheck>
    <disabled>no</disabled>
    <scan_on_start>yes</scan_on_start>
    <frequency>3600</frequency> <!-- shortened for testing -->
    <directories check_all="yes" realtime="yes">/home/hunter/Documents</director
ies>
    <directories check_all="yes" realtime="yes">/etc</directories>
    <ignore type="sregex">.log$|.swp$</ignore>
    <nodiff>/etc/ssl/private.key</nodiff>
    <skip_nfs>yes</skip_nfs>
    <skip_dev>yes</skip_dev>
    <skip_proc>yes</skip_proc>
    <skip_sys>yes</skip_sys>
    <process_priority>10</process_priority>
    <max_eps>50</max_eps>
    <synchronization>
      <enabled>yes</enabled>
      <interval>5m</interval>
      <max_eps>10</max_eps>
    </synchronization>
  </syscheck>
```

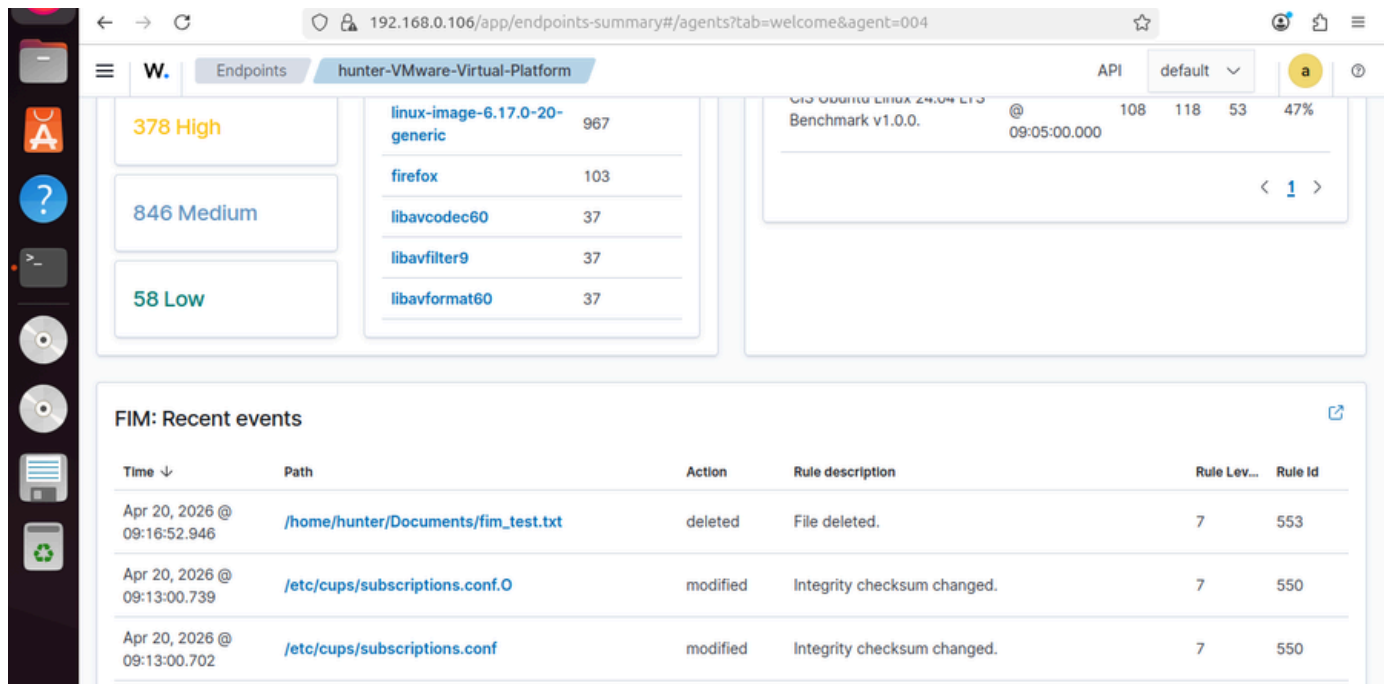
Screenshots

Screenshot 02 : File integrity Alerts.



Screenshots

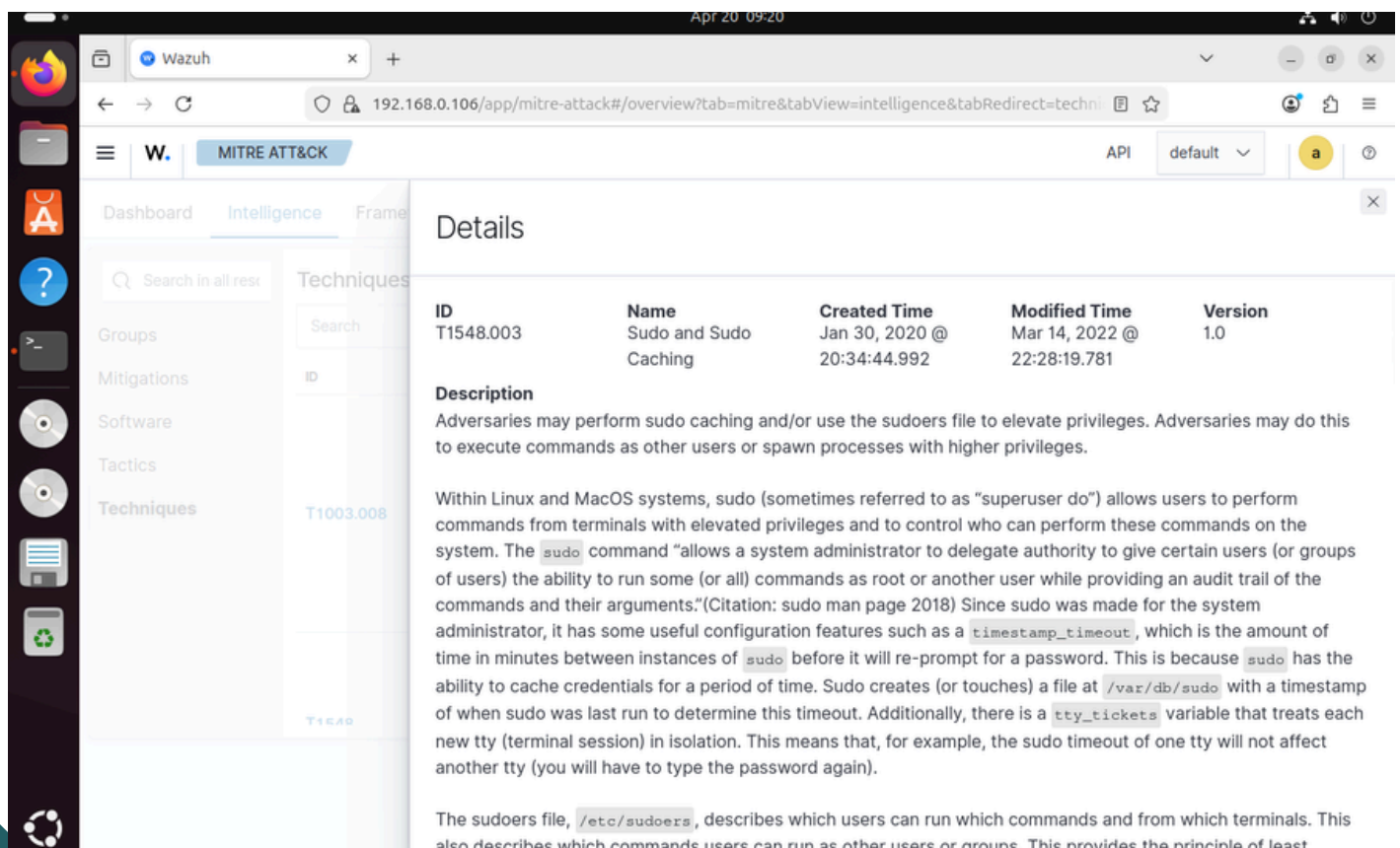
Screenshot 03 : File Modification Event.



The screenshot shows the Wazuh dashboard interface. The top navigation bar includes 'Endpoints' and 'hunter-VMware-Virtual-Platform'. The main content area displays 'FIM: Recent events' with a table of file modification events.

Time ↓	Path	Action	Rule description	Rule Lev...	Rule Id
Apr 20, 2026 @ 09:16:52.946	/home/hunter/Documents/fim_test.txt	deleted	File deleted.	7	553
Apr 20, 2026 @ 09:13:00.739	/etc/cups/subscriptions.conf.O	modified	Integrity checksum changed.	7	550
Apr 20, 2026 @ 09:13:00.702	/etc/cups/subscriptions.conf	modified	Integrity checksum changed.	7	550

Screenshot 04 : Vulnerability Alert Filtered by Rule ID.



The screenshot shows the Wazuh dashboard interface. The top navigation bar includes 'MITRE ATT&CK'. The main content area displays 'Details' for a vulnerability alert, filtered by Rule ID T1548.003.

ID	Name	Created Time	Modified Time	Version
T1548.003	Sudo and Sudo Caching	Jan 30, 2020 @ 20:34:44.992	Mar 14, 2022 @ 22:28:19.781	1.0

Description

Adversaries may perform sudo caching and/or use the sudoers file to elevate privileges. Adversaries may do this to execute commands as other users or spawn processes with higher privileges.

Within Linux and MacOS systems, sudo (sometimes referred to as "superuser do") allows users to perform commands from terminals with elevated privileges and to control who can perform these commands on the system. The `sudo` command "allows a system administrator to delegate authority to give certain users (or groups of users) the ability to run some (or all) commands as root or another user while providing an audit trail of the commands and their arguments." (Citation: sudo man page 2018) Since sudo was made for the system administrator, it has some useful configuration features such as a `timestamp_timeout`, which is the amount of time in minutes between instances of `sudo` before it will re-prompt for a password. This is because `sudo` has the ability to cache credentials for a period of time. Sudo creates (or touches) a file at `/var/db/sudo` with a timestamp of when sudo was last run to determine this timeout. Additionally, there is a `tty_tickets` variable that treats each new tty (terminal session) in isolation. This means that, for example, the sudo timeout of one tty will not affect another tty (you will have to type the password again).

The sudoers file, `/etc/sudoers`, describes which users can run which commands and from which terminals. This also describes which commands users can run as other users or groups. This provides the principle of least

Screenshots

Screenshot 05 : Vulnerability mitigation tactics according to NIST-800 Framework.

The screenshot displays the MITRE ATT&CK framework interface. The left sidebar contains navigation links: Dashboard, Intelligence, Framework, Groups, Mitigations, Software, Tactics, and Techniques. The main content area is titled 'Details' and shows a table of mitigation tactics. The table has columns for ID, Name, and Description. The first row is 'M1022 Restrict File and Directory Permissions'. The second row is 'M1026 Privileged Account Management'. The third row is 'M1028 Operating System Configuration'. The table is filtered by 'T1003.008' and 'T1040'. The interface also includes a search bar, a 'Rows per page: 5' dropdown, and a pagination control showing '1'.

ID	Name	Description
No items found		
Mitigations		
ID	Name	Description
M1022	Restrict File and Directory Permissions	Restrict access by setting directory and file permissions that are not specific to users or privileged accounts.
M1026	Privileged Account Management	Manage the creation, modification, use, and permissions associated to privileged accounts, including SYSTEM and root.
M1028	Operating System Configuration	Make configuration changes related to the operating system or a common feature of the operating system that result in system hardening against techniques.

Rows per page: 5

< 1 >

Software

03 Challenges & Solutions

- Challenge : No alerts appearing immediately after changes.
Solution: Restarted Wazuh agent and waited for the next syscheck scan cycle.
- Challenge: High volume of alerts.
Solution: Applied dashboard filters and tuned FIM settings (<frequency>)
- Challenge : Understanding specific rule IDs and Alerts.
Solution : Reviewed wazuh recommended tactics and documentetion of wazuh rules.

05. SOC Skills Gained

- File Integrity Monitoring (FIM) configuration and management
- Alert triage and investigation in Wazuh dashboard
- Basic incident investigation and documentation
- Understanding how everyday system changes generate security events